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Received: 29 November 2024 | Accepted: 30 March 2025

DOI: 10.1111/bjet.13586

ORIGINAL ARTICLE

British Journal of Educational Technology BERA

Development of a generative AI-powered teachable agent for middle school mathematics learning: A design-based research study

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This paper reports on a design-based research (DBR) study that aims to devise an artificial intelligence (AI)-powered teachable agent that supports secondary school students' learning-by-teaching practices of mathematics learning content. A long-standing pedagogical practice of learning-by-teaching is powered by a recent advancement of generative AI technologies, yielding our teachable agent called *ALTER-Math*. This study chronicles one usability testing and three cycles of iterative design and implementation process of *ALTER-Math*. The three empirical studies involved a total of 320 middle school students and six teachers in authentic classroom settings. The first study was exploratory, focusing on the qualitative feedback from the students and teachers through open-ended surveys, interviews and classroom observations. The second study yielded a medium-high ($M = 3.26$) quantitative survey result on students' perceived engagement and usability on top of the qualitative findings. Finally, the final study included pre- and post-knowledge tests in a quasi-experimental study design as well as student and teacher interviews. The final study revealed a bigger significant knowledge improvement in students who used *ALTER-Math* compared to the control group, suggesting a positive impact of AI-powered teachable agents on students' learning. The design implications learned from multiple iterations are discussed to inform the future design of AI-powered learning technologies.

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KEYWORDS
AI-assisted learning environment, generative AI, learning-by-teaching, mathematics learning, teachable agent

Practitioner notes

What is already known about this topic

- Learning-by-teaching is a long-standing effective pedagogical strategy to enhance students' domain knowledge and feelings of responsibility in learning.
- Various teachable agents have been developed and have demonstrated benefits in students' learning.
- Generative AI offers the potential to provide naturalistic, contextualised and adaptive conversations.

What this paper adds

- Develops a novel generative AI-powered teachable agent for middle school mathematics learning, called *ALTER-Math*.
- Reports the iterative design process involving empirical classroom implementations of *ALTER-Math*.
- Reveals a bigger significant improvement in the student's mathematical knowledge after using *ALTER-Math*, compared to the control group.

Implications for practice and/or policy

- Researchers can be inspired by this design example of a theoretically grounded generative AI learning technology.
- Educational technology designers could hear the real voices of students and teachers about the generative AI learning technologies.
- Researchers and educational technology designers could be directed by the design implications to the future design of AI-powered learning technologies and teachable agents.

INTRODUCTION

It is a long-standing belief that people learn better when they teach others (Frager & Stern, 1970). This empirical belief has been developed into a widely accepted pedagogical practice, namely 'learning-by-teaching' (Duran, 2017). Learning-by-teaching has been identified as an effective way to promote a learner-centred learning environment with a higher level of learner responsibilities (Chase et al., 2009), enhancing deep knowledge construction (Fantuzzo et al., 1992), self-efficacy and confidence in learning (Roscoe & Chi, 2007). A body of research has reported positive effects of learning-by-teaching practices in different settings, such as peer tutoring (Roscoe & Chi, 2007), collaborative learning (Kafai, 1991) and reciprocal learning (Palinscar & Brown, 1984). In addition to these human-to-human interactions (eg, student-to-student interaction in peer tutoring), studies have found that the pedagogical effects of learning-by-teaching are also found in the virtual learning environment, such as teaching fictitious peers or computer systems (Fiorella & Mayer, 2013). In turn, researchers and educators have created virtual learning-by-teaching environments called teachable agents (Song, 2017). A teachable agent is a type of intelligent tutoring system that

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simulates a less-knowledgeable peer to convert the role of learners to that of teachers (Zhao et al., 2012). With the advancement of educational technologies, various types of teachable agents have been designed and developed for different target learners in different contexts (eg, Biswas et al., 2016; Matsuda et al., 2015; Song, 2017, see "Teachable Agent" Section for details). However, those teachable agents to date have faced technical challenges in processing and generating natural languages to allow students to have human-like interactions out of the pre-defined options (ie, clicking the button to respond to the agent).

The recent advancement of generative AI has offered the opportunity to transform the design and development of teachable agents. The use of large language models can simulate far more naturalistic conversations with humans (Aher et al., 2022) and adapt to the learner's levels, interests, and learning preferences (Vlachos et al., 2023). The use of generative AI for teachable agents would provide far more naturalistic, contextualised and personalised learning-by-teaching experiences. Motivated by this identified potential, this study designed and developed a generative AI-powered teachable agent to support secondary school students' mathematics learning, theoretically grounded in Social Constructivism and Behaviourism (eg, learning-by-teaching, contrasting cases, social psychological intervention and gamification). This paper reports a design-based research process of three classroom studies involving 320 middle school students by providing them with user experiences of our developed technology in authentic middle school classrooms. We name the generative AI-powered teachable agent for mathematics learning *ALTER-Math*. This study specifically sought to answer the following research questions (RQs).

- RQ1. How is *ALTER-Math* perceived by middle school students over time in an iterative design process?
- RQ2. How does the use of *ALTER-Math* impact middle school students' mathematics knowledge?
- RQ3. What are the design implications for an effective design of AI teachable agents for middle school mathematics learning?

LITERATURE REVIEW

AI-assisted learning environment

AI-assisted learning environments are educational settings where AI tools, such as intelligent tutoring systems (ITS), adaptive learning platforms and AI virtual agents, are used to support and enhance personalised learning and teaching processes (Laato et al., 2023; Seo et al., 2021). These environments can be implemented in traditional classrooms, online platforms and blended learning settings. Previous research indicates the potential of AI technologies to facilitate individualised learning experiences in various educational settings, including K-12 education, higher education and professional training programs (Lin et al., 2023; Xu & Ouyang, 2022a). For example, researchers have developed an interactive ITS for Physics education, featuring a virtual agent that taught physics concepts and used decision algorithms to adjust the agent's level of support (Myneni et al., 2013). Ledesma and García (2017) developed a recommendation system for math problems tailored to each student's individual levels and behaviours. Hooshyar et al. (2018) introduced an innovative flowchart-based ITS, employing Bayesian network techniques to replicate human instructor-led, one-on-one instruction. These tools and applications focus on providing personalised and real-time guidance and feedback and help students in the learning process, reducing the teaching load of instructors.

The utility of AI has also been used to enhance communication and support between students and instructors in the form of teaching assistants (Goel & Polepeddi, 2018). These assistants facilitate more dynamic and responsive interactions, effectively bridging communication gaps that often exist in traditional educational settings, where instructors use one-to-many instruction, which limits personalised interaction between teachers and students. Additionally, the implementation of AI scoring systems (Seo et al., 2021) plays a crucial role in this enhancement, offering more personalised and adaptive assessments of student performance and relieving teachers' burdens. These ways of using AI as an instructor or teaching assistant have also raised the concern that AI might replace the original role of instructor and work as a new subject to individually convey knowledge (Xu & Ouyang, 2022a, 2022b). While AI has been increasingly used to provide intelligent tutoring, such as personalised instruction (eg, Garcia-Varela et al., 2025; Khan, 2019) or support teachers' instruction as teaching assistants (eg, Goel & Polepeddi, 2018; Song et al, 2025b), there is limited research on how AI can be used for students to teach, promoting learning-by-teaching practices.

Teachable agent

A teachable agent is a type of intelligent tutoring system that simulates a one-on-one tutoring experience based on learning by teaching theory, making learning more engaging and effective (Zhao et al., 2012). Teachable agents are typically embedded within an interactive chatbot learning environment, which allows students to engage with the agent by inputting information, asking the agent to perform tasks or even assessing the agent's understanding of the taught material (Baranwal, 2022). The agent's responses and actions are based on AI algorithms that simulate learning, making mistakes and gradually improving over time, much like a human learner (Lee et al., 2023; Weisz et al., 2019). This allows students to practice in authentic contexts and receive timely, relevant explanations (Sinclair et al., 2019).

A critical component of teachable agents is the feedback mechanism (Leelawong et al., 2002; Silvervarg et al., 2021). As students teach the agent, the system provides feedback on the agent's performance, which indirectly reflects the student's teaching effectiveness and understanding of the subject matter (Johnson & Lester, 2018). This feedback can take various forms, such as quizzes, simulations or real-world tasks that the agent attempts to complete based on the instruction it has received (Baranwal, 2022; Matsuda et al., 2015). For example, teachable agents can examine students' responses and provide personalised feedback, helping students recognise their strengths and areas for improvement (Silvervarg et al., 2021). This promotes self-regulated learning by involving students in the active teaching of the agent, naturally leading them to organise, monitor and evaluate their own learning progress (Silvervarg et al., 2021). Additionally, such an interactive aspect of the teachable agent also boosts students' enthusiasm and interest in the subject matter (Baranwal, 2022).

SimStudent (Matsuda et al., 2007, 2010, 2015) is an example of a teachable agent used in the domain of mathematics that allows students to teach how to solve algebraic equations. The process of instructing SimStudent helps students to better understand the rules and steps involved in solving equations, as they must articulate these processes clearly for the agent to learn (Matsuda et al., 2013). As students correct SimStudent's mistakes, they reinforce their own knowledge and problem-solving skills (Matsuda et al., 2015). Some educational platforms offer virtual math coaches, which are AI agents that students can interact with to learn math. For example, Squirrel AI Learning's virtual math coaches assess students' knowledge gaps, deliver customised content and engage students in solving math problems (Li et al., 2018). They encourage students to teach back the concept to the AI, enhancing their understanding and retention of mathematical concepts. Zearn is another virtual math coach that offers an interactive experience where students solve math problems and

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explain reasoning processes, thus improving their mathematical understanding (Morrison et al., 2019). Finally, Betty's Brain was another teachable agent for middle school science learning that provides a visual concept mapping of Betty as Betty learns from the student (Biswas et al., 2016).

The aforementioned examples of teachable agents can be programmed to learn from the student's teaching, asking questions and posing problems that require the student to explain concepts and methods. However, they focus more on utilising adaptive learning algorithms to adjust the difficulty and sequence of existing content and rely on predefined or static learning material libraries, which could limit the diversity and richness of the learning content. In addition, traditional teachable agents employed simple interactions, such as button-clicking, to choose the response options (eg, SimStudent, Betty's Brain), which could prevent naturalistic and social learning-by-teaching interactions. The recent advancement of generative AI offers the potential to equip teachable agents with more adaptive and personalised learning support, as well as a more human-like and natural conversation. Therefore, this study focuses on the design and practical usage of generative AI for teachable agents.

THEORETICAL FRAMEWORK

The design of *ALTER-Math* is grounded in the theoretical framework informed by learning theories of Social Constructivism and Behaviourism. The theoretical framework consists of learning-by-teaching, contrasting cases, social psychological intervention and gamification.

Learning-by-teaching

Learning by teaching is a Social Constructivist's theory that maintains students can learn more effectively when they teach others (Fiorella & Mayer, 2014; Frager & Stern, 1970; Roscoe & Chi, 2008). Learning-by-teaching is realised in different formats, such as peer tutoring, where students engage in one-on-one or one-on-multiple tutoring sessions to teach younger or less knowledgeable peers (Roscoe & Chi, 2007); collaborative learning, where students naturally teach each other (Kafai, 1991); students' role-playing of acting as a teacher and presenting teaching materials in the classroom (Duran, 2017); or students teaching a digital system that simulates a less knowledgeable peer, which is a teachable agent (Biswas et al., 2016). Research has demonstrated the benefits of learning-by-teaching practices, such as enhancing students' deep understanding of domain knowledge (Fantuzzo et al., 1992), improving social and communication skills (Roscoe & Chi, 2007), and developing students' feelings of responsibility in learning (Chase et al., 2009).

Despite the general understanding of the effects of learning-by-teaching, researchers have been interested in 'what really explains the learning potential offered by teaching' (Duran, 2017, p. 3). Bargh and Schul (1980) framed the process of learning-by-teaching into three phases: (a) preparing for teaching (ie, learning to teach (Duran, 2017)), (b) presenting the teaching materials to the tutees [ie, learning and explaining (Duran, 2017)], (c) interacting with the tutees by question-and-answer and going over any confusing parts of the teaching materials [ie, teaching by interacting (Duran, 2017)]. Research has been interested in investigating which phase(s) among these three could be critical for successful learning-by-teaching (Duran, 2017). First, the effect of teaching expectancy (ie, the expectancy of having to teach when they learn) was investigated in different contexts. For example, Fiorella and Mayer (2014) examined the effects of 'preparing to teach' compared to 'preparing and actually teaching'. Their study found that the 'preparing and actually teaching' group outperformed the 'only preparation' and 'non-learning-by-teaching' group (control group) in both

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long- and short-term periods; 'only preparing' was more effective than the control group; however, they could not retain the knowledge as much as the 'actual teaching' group, when tested a week later (Fiorella & Mayer, 2014). In addition, Lachner et al. (2022) disentangled the effect of interactions in a learning-by-teaching context by comparing interactive teaching and non-interactive teaching. The suggested effects of non-interactive teaching practices were argued to have stemmed from phenomena such as retrieval practice (ie, retrieving previously learned knowledge to teach), generative processing (ie, generating an explanation of the teaching material), and social presence (ie, being aware that 'real' persons exist in the computer-mediated learning environments) (Lachner et al., 2021). Interestingly, the effectiveness of learning-by-teaching was still valid without the presence of the actual human audience but with fictitious or imagined peers (Fiorella & Mayer, 2013; Hoogerheide et al., 2016; Lachner et al., 2021).

The review of the literature suggests important elements of learning-by-teaching to improve students' deeper learning of knowledge and enhanced learning experiences. The teachable agent system proposed in this study aims to provide a learning experience that includes various aspects of learning-by-teaching practices, including preparing to teach that promotes retrieval of knowledge, actual teaching to encourage generative knowledge processing and interacting with the (digital) tutee that supports the feeling of social presence.

Contrasting cases

Contrasting cases integrate both exemplary (positive) and flawed or incorrect (negative) examples, fostering a deeper appreciation of the key distinctions between high-quality and subpar responses or solutions (Chin et al., 2016; Schwartz et al., 2011; Schwartz & Martin, 2004). Contrasting cases have been used in subjects such as mathematics and physics, showing positive effects on learning. For example, Chin et al. (2016) revealed that contrasting cases led students to pay extra attention to the discreteness of the phenomena and solutions of physics. Similarly, Sidney et al. (2015) revealed that the contrasting cases method improved the quality of self-explanation of mathematics among undergraduate students. It can be explained that contrasting cases effectively counter students' tendency to overgeneralise by highlighting that mathematical rules and principles have their exceptions and are not universally applicable (Schwartz et al., 2011). Moreover, contrasting cases enrich the learning-by-teaching strategy by offering a broad and varied collection of examples for students to instruct and elucidate, encouraging a more critical examination and comprehension of the subtle differences and practical applications of various mathematical concepts.

Social psychological intervention and gamification

Social psychological interventions are strategies designed to enhance outcomes by focusing on psychological processes—such as perceptions, beliefs and attitudes—instead of external conditions (Yeager & Walton, 2011). In a meta-analysis on social psychological interventions' effects on learning outcomes in the post-secondary context, a medium effect size was reported ($d = 0.47$) (Lazowski & Hulleman, 2016). Empirically, Borman et al. (2021) conducted a large-scale study examining the long-term effects of a brief self-affirmation intervention (ie, reflection on other meaningful aspects of their personal identities outside of academics) designed to mitigate stereotype threat among seventh-grade African American and Latino students. Their findings revealed that the intervention reduced the achievement gap by 50% annually and increased on-time graduation rates

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for treated minority students by 10 percentage points. In another example, Samuel and Warner (2021) explored the impact of combined mindfulness and positive psychology intervention embedded in a developmental statistics course, finding that it significantly reduced math anxiety and increased math self-efficacy among community college students. In our project, students are empowered through values affirmation and confidence building by engaging in instructional roles with a novice AI teachable agent, designed to inspire and celebrate their efforts, fostering a positive and supportive learning experience regardless of their prior knowledge.

In relation to social psychological intervention, the design of *ALTER-Math* employs the elements of gamification to enhance students' engagement, motivation and attention in learning. Gamification is the use of game design elements, such as points, progress, leaderboards, badges, competitions or missions in non-game contexts (Deterding et al., 2011). Gamification has been applied to learning, such as the design of instructional programs or educational technologies (Sailer & Homner, 2020). Sailer and Homner (2020) reported significant small effects of gamification on students' cognitive, motivational, and behavioural learning outcomes through a meta-analysis. Some studies investigated the effects of specific elements of gamification, such as points (Attali & Arieli-Attali, 2015), competition (Ho et al., 2022), badges, and leaderboards (Huang & Hew, 2015). Attali and Arieli-Attali (2015) found a positive effect of points in students' engagement, perceived efforts put in learning and likeness among middle school students. Ho et al. (2022) suggested mixed effects of competition in learning, including Burguillo (2010) reporting benefits of competition in learning, such as improved engagement and performance, as opposed to the negative effects, such as anxiety and lowering students' self-esteem (Lam et al., 2004). Huang and Hew (2015) report positive effects of points, badges and leaderboards in improving students' motivation in online learning. Based on the review of gamification, we included the game elements in the design of *ALTER-Math* to increase students' cognitive, motivational and behavioural learning outcomes.

METHODS

Design-based research methodology

We adopted the design-based research (DBR) methodology to develop the AI-powered teachable agent learning tool for secondary school students. DBR is a systematic and flexible methodology to improve educational practices through an iterative process in collaboration among researchers and practitioners in authentic educational settings (Wang & Hannafin, 2005). DBR utilises interventions to introduce new educational methods and tools and seek the possibilities to 'improve' them (Forte, 2009). DBR mostly involves multiple iterative design and development cycles and pursues progressive refinement (Collins et al., 2016). In DBR studies, both effective and ineffective features of the interventions are appreciated as they both provide possibilities to improve in the next design phase (Li & Chu, 2018). In this research, we conducted an iterative design study, including one teacher interview and three classroom studies to improve the design of *ALTER-Math*.

Research design

Based on the spirit of progressive refinement of design-based research, we set up an iterative research design involving four steps inspired by Schmitz et al. (2015) and Song and Lim (2021). Figure 1 depicts the process of our research design consisting of four

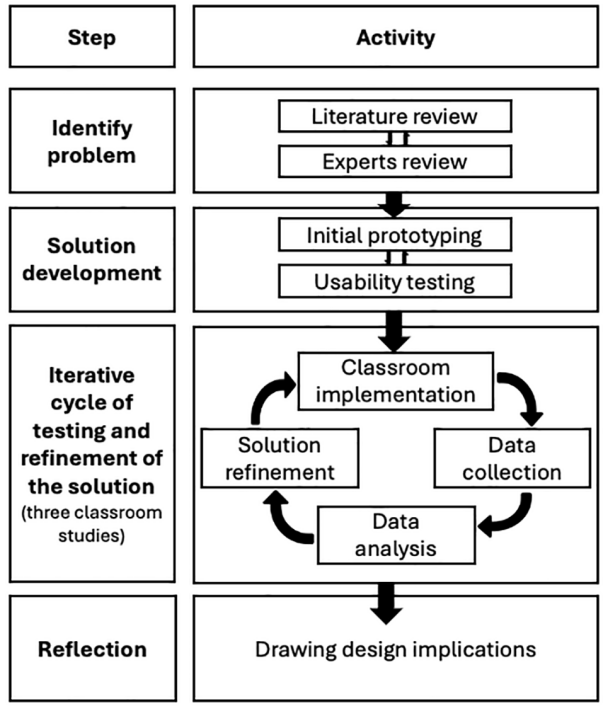


FIGURE 1 Research design: Steps and activities.

steps and activities in each step. The steps include (1) problem identification through literature review and expert reviews, (2) solution development including initial prototyping and usability testing, (3) an iterative cycle of testing and refinement of the solution (prototype), which includes three rounds of classroom implementations and (4) reflection that involves drawing design implications. In the first step of problem identification, we reviewed the related literature on the theories and practices of AI-assisted learning, learning by teaching, and teachable agents, as described in "Literature Review" Section. After we identified the initial design principles and prototype ideas from the literature, we discussed these with five experts in AI education, learning sciences, instructional design and educational technology, who are faculty members at Research 1 universities in the United States. Based on these discussions, we developed the initial prototype. The initial prototype was the minimal viable product equipped with basic functions such as enabling the AI agent to learn from historical contexts between AI and specific students, enhancing socio-emotional support using few-shot learning prompts with illustrative examples, and preventing potential conversational digression through a rule-based limit of five turns per interaction. The prototype was used for usability testing and involved five in-service middle school teachers. The usability testing was conducted to ensure there are no critical usability problems that could hamper students' learning experiences from teachers' perspectives through teacher interviews. The third step involves iterative cycles of testing and refinement of the solution, with three classroom studies. The first and second classroom studies involved the same two 6th-grade teachers and their classes of 140 students. The purpose of those studies was to investigate middle school students' experiences with the usability of the developed tool and explore their perceptions of what was good, not good and what needs to be improved about *ALTER-Math*. The final classroom study expanded participation to 212 students and adopted a quasi-experimental study to examine the impacts of *ALTER-Math* on students' mathematical knowledge. Finally,

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our research team gathered and reflected on our experiences and identified the design implications for further improvement of *ALTER-Math*.

DESIGN AND DEVELOPMENT OF *ALTER-Math*

Initial prototyping

The AI teachable agent's design is rooted in three pedagogical strategies shown to enhance student engagement, foster learning ownership, and deepen understanding of mathematics: learning-by-teaching, contrasting cases and social psychological interventions (see "Theoretical Framework" Section). Shifting students' role to instructors to AI requires them to articulate their knowledge and understanding, which can promote cognitive engagement and facilitate their conceptual understanding of content (Biswas et al., 2005; Chase et al., 2009). The paradigm shift from reactive to proactive learning is enabled by the integration of large language models (LLMs) and retrieval-augmented generation (RAG) (see Figure 2a). RAG serves to improve LLMs' ability to produce responses that align with human expectations by leveraging external databases for contextual information, thereby generating more pertinent and contextually relevant outputs. (Levonian et al., 2023; Lewis et al., 2020). We employed OpenAI's GPT-4-Turbo for generating content, specifically prompting the LLM to simulate a middle school student struggling with mathematics. Additionally, we utilised RAG to supply context on potential misconceptions or common mistakes related to the chosen math topic by students, drawing from a comprehensive database of over three million genuine math Q&As from Math Nation, an online math learning platform actively adopted by over one million students annually.

The second set of features is guided by social psychological interventions. Our solution enables students to interact with a teachable agent engineered with personalities

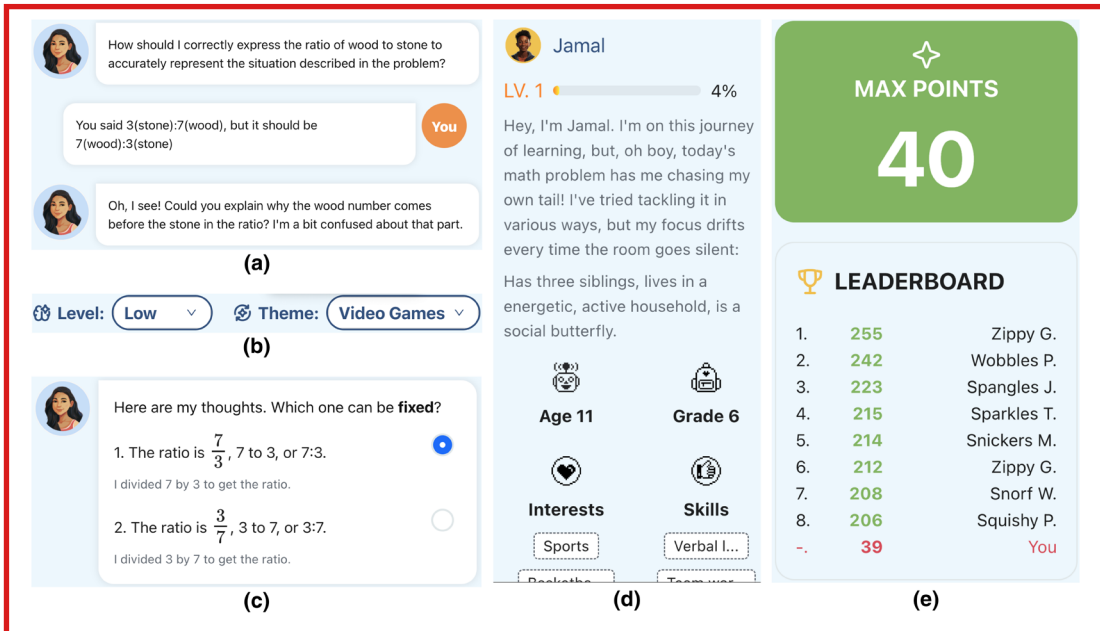


FIGURE 2 Selected features of the AI teachable agent. (a) LLM-powered learning by teaching. (b) Learning with choices. (c) Contrasting cases. (d) Persona-engineered psychological interventions. (e) Gamification.

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(see Figure 2d) to intrigue, empower, and reassure their learning as instructors via few-shot prompting by providing examples of desirable conversational interactions (Brown et al., 2020). Meanwhile, students are given the choices to personalise their teaching experience by selecting math problems of three difficulty levels and nine themes (eg, influencers, environmental issues, music) that resonate with their interests (see Figure 2b). We conducted an experimental investigation to facilitate this ownership of learning using chain-of-thought (CoT, eg, instructing LLMs to break down its reasoning process into intermediate steps before arriving at a final answer following a fixed set of examples, Wei et al., 2022), active-prompt (eg, similar to CoT but with dynamically selected examples, Diao et al., 2023), and tree-of-thought prompt engineering techniques (ToT, branching out intermediate steps and traversing based on evaluation results, Yao et al., 2023). Examples of these prompts and the vetting interface can be found in Appendix A (Figure A1). In the experiment, three content experts with PhD degrees in math education were randomly assigned with 350 GPT-4 generated problems using the three techniques, respectively. The goal was to evaluate the generated problems' standards-alignment, context relevance and problem accuracy. The examination showed that CoT could achieve the best evaluation performance, with 79.2% approval and 8.7% minor edits (<5% content revision) rates.

Finally, students' learning-by-teaching with the AI teachable agent is enacted with contrasting cases (see Figure 2c). We generated contrasting cases using LLMs together with the learning choices, which were vetted by professionally trained experts in mathematics. To further engage students' participation in learning-by-teaching with contrasting cases, we also gamified their teaching process by introducing an experience point system and leaderboard (see Figure 2e). Students' teachable agent would gain different points based on their selected difficulty level and types of interaction (eg, making a choice, typing instruction, revising instruction). Students can also compare their accumulated points to peers through a leaderboard, understanding others' engagement and accomplishment.

Usability testing with teachers

We conducted usability testing of the prototype to ensure that the AI-teachable agent had no major usability problems that could impair student experience and performance (Torres, 2001). Five K-12 teachers participated in semi-structured online interviews. To ensure the representativeness of the participants' sample, we used the stratified sampling strategy (Robinson, 2014) based on gender and school location when sending out invitations. Among all participants, three were females, two were males and all had more than 5 years of teaching experience; two interviewees were from economically disadvantaged districts. The duration of each interview ranged from 40 to 50 minutes. Before the interviews, all participants consented to participate in the study.

During the first part of the interview, we asked participants their general opinions on learning by teaching using AI, answering the following questions: “Which do you prefer and why: letting students teach AI versus letting AI teach students?”. Next, we went through each feature of the prototype and asked them to try the prototype using the following prompt: “Pretend you are a student using this tool. Please ‘think aloud’ while you are using this tool and let us know your thoughts on each part of this tool”. Finally, we asked about their experience through the following questions: “Do you think this tool, in general, supports your students learning math?” and “What are your concerns on this tool for your students?”. All interviews were audio-recorded and transcribed afterward through a reputable transcription service before the qualitative data analysis. A qualitative data analysis software, Atlas.ti, was used to code and synthesise data based on our research questions. Following McDonald et al. (2019)'s methods and given the direct alignment between the study protocol and the analysis themes, statements of interest

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were jointly coded by two researchers according to the pre-set themes of the study protocol: (a) teachers' general opinions on letting students teach AI; (b) teachers' experience as a user on each feature of the prototype; (c) teachers' opinions on potential benefits and expected concerts of the AI teachable agent. The authors examined the resulting coded statements and had several discussions to select instances illustrating the teachers' perceptions.

First, most teachers (four out of five) do not have a tendency toward 'teaching AI' or 'letting AI teach' but acknowledged the potential benefits of AI to increase students' interest ("*I think both are attractive to students; you know, AI can increase their interest in learning*"). However, teachers also expressed their concerns, especially the two teachers from economically disadvantaged districts, who stated that their students may not be able to teach someone as they lack the expression skills. Second, the main features of the prototype worked smoothly when teachers went through the whole process of using the prototype step-by-step. They talked about some specific suggestions, such as some confusing prompts for students. For example, when the system asks the students to choose the solution that needs to be improved among the contrasting cases, the prompt says, 'Which one can be improved?'; teachers suggested changing this to be clearer, so we changed it to be 'Which one can be fixed?' with bold text. In addition, the accessibility issues of reading the texts were pointed out. Thus, we added a text-to-speech feature to the agent's utterances to help lessen the reading barrier for some students. Finally, three of the five teachers also suggested having a teacher dashboard to enable teachers to know more clearly about students' behaviours and performance using this tool. All these suggestions were discussed after the study and were addressed in the next prototype. Overall, all teachers expressed that the current prototype would be good to use for their students after addressing the usability issues mentioned above and showed their excitement about the use of *ALTER-Math* in their math classes.

ITERATIVE DESIGN CYCLES

First classroom study

Classroom context and participants

The first classroom study was conducted in December 2023. The participants were 108 6th-grade students in six separate class periods of 20–25 students in a public school in the Southern part of the United States. Two middle school math teachers (one male and one female) were in charge of these students. The school is a research school affiliated with a research-based (R1) university. The gender breakdown included 50 boys, 51 girls and 7 identifying as other. The ethnic composition was 29 Black/African American, 56 White, 15 Hispanic, 5 Asian and 3 others. Ninety-four speak English as their first language, 4 speak Spanish as their first language, and 22 are multilingual. In addition, we hired four facilitators who are graduate students majoring in Educational Technology, General Education and Math Education.

Prior to using the teachable agent, students learned the contents with their teachers while they were informed that they would be teaching this learned content to others later. This phase was designed to provide students with a teaching expectancy to increase the potential effects of learning by teaching. We accommodated the learning unit to the classes, covering the unit of Rate and Ratios, which aligns with the Florida state's 6th-grade benchmarks of MA.6.AR.3.5., "Solve mathematical and real-world problems involving ratios, rates, and unit rates." We used AI to generate eight problems on diverse real-world topics (eg, video games, YouTube) and difficulty levels (ie, high, medium, low). The AI-generated problems went through expert reviews by three content experts with Ph.D. degrees in Math Education

to check whether they align with the standards and whether the content is accurate, safe and readable. The first study only took one class period of 42 minutes. Two facilitators and one teacher were present during the class periods and provided guidance, instructions and feedback to the students in each of the two classrooms. While most conversations between students and the teachable agent stayed as math-related topics, some conversations went off-topic. To address this, we included the guidance in our prompt to redirect the students if they go off-topic (see Appendix B for more details). Figure 3 illustrates students who are engaging in learning by teaching math to the AI agents.

Data collection and analysis

We collected qualitative data from open-ended surveys, interviews, and classroom observations. First, the survey was conducted on all student participants after the classroom implementation. The survey link was distributed during the in-person class.[1] The first four questions are about students' demographics (ie, gender, race, first language and other languages they speak). The questions on the languages were asked to learn about their language background as it could potentially impact students' learning experiences with *ALTER-Math* that primarily used English. Following that, we included two open-ended questions to learn about students' opinions on (1) what they liked about *ALTER-Math* and (2) what they did NOT like about *ALTER-Math* and how it can be improved.

After the classroom implementation, we conducted interviews with teachers and students separately. First, we interviewed the two teachers about their experiences using *ALTER-Math* in their classrooms and their observations of the students using the tool. Additionally, we aimed to gain insights about the teacher dashboard we were trying to develop based on the previous usability interview with five in-service teachers (see "Usability Testing with Teachers" Section). The interview was conducted by two graduate researchers, recorded and transcribed using a professional online transcription service[2] afterward. The interview prompts are outlined in Table 1. Second, a subset of students participated in semi-structured focus group interviews on the day following their experience to provide detailed and diverse insights into their experience of the AI teachable agent in math learning (Vaughn et al., 1996). To find optimal interviewees, facilitators, and teachers took notes of students of interest based on their engagement, interest, and communication skills during the classroom implementation. Among the students listed as suggested, a total of 14 students volunteered to participate in the interviews.



FIGURE 3 Students are engaged in learning with *ALTER-Math*.

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TABLE 1 Teacher interview prompt.

1. How was your experience with using *ALTER-Math* in your classroom?
2. What were the major benefits of using *ALTER-Math*? How did the students react?
3. How engaged do you think the students were compared to the traditional class?
4. What were the major challenges and difficulties of using *ALTER-Math*? How would you improve it?
5. Would you want to use *ALTER-Math* in your classroom in the future?
6 If we provide a teacher dashboard, what kinds of information would you want to know as a teacher?

TABLE 2 Student interview prompt.

1. So far, how has your experience been with *ALTER-Math*?
2. What did you like about *ALTER-Math*?
2.1. Was it fun to interact with alter math? Why? Why not?
3. How was the onboarding process to *ALTER-Math*? Did the tutorial help? Why, why not?
3.1. What features in *ALTER-Math* have been easy/difficult to understand?
3.2. Probe: How would you change it to improve it?
4. How does interacting with *ALTER-Math* help you understand math concepts?
4.1. What math concepts did you teach? Did you learn anything new?
4.2. What were the differences between learning as a student and learning by teaching acting in the teacher's role?

Three interview groups were formed, consisting of 4–6 students and led by one graduate researcher, respectively. The duration of these interview recordings ranged from 17 to 29 minutes, averaging 25 minutes. The three researchers shared an interview prompt (Table 2) consisting of nine questions about their experiences with *ALTER-Math* and had the flexibility to ask follow-up questions if needed.

Finally, we collected qualitative observation notes from teachers and researchers who were present in the classroom implementations. One teacher and two facilitators were present in each learning session, working as observers. They took notes about any technical errors, confusing points students expressed about the interface design, prompt design, and interactive use of the tool, common questions students asked (eg, difficult words or terminology), and any other struggles students were having while using *ALTER-Math*. The observation notes served as a triangulation of the other main data sources to enhance the credibility of the data analysis.

In the first study, we mainly collected qualitative data from students' responses to the open-ended survey questions and semi-structured focus-group interviews after the classroom study. In addition, we collected teachers' opinions from the interviews and the observation notes from facilitators and teachers during the classroom activities. We conducted a thematic analysis to examine all qualitative data since it aligned with our research objectives and the nature of the data obtained. Thematic analysis was chosen because it is effective in exploratory studies for identifying, articulating, and interpreting participants' experiences concerning specific issues and prevailing practices (Braun et al., 2016), especially with various data sources, including interviews, open-ended responses and observations (Howitt, 2019). Our analysis employed a combined inductive and deductive approach to thematic analysis (Smith, 2015). Initially, two researchers independently carried out a deductive analysis to formulate initial themes, drawing on their knowledge of existing literature, theoretical frameworks, and their insights into Artificial Intelligence in Education (AIED). Subsequently, an inductive analysis was performed by the same researchers to discover

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codes and themes that were not predefined. This was followed by an iterative process of reviewing all transcribed data, during which themes were categorised, refined and clarified to ensure their uniqueness and the clear interrelationships among them. This process continued until a consensus was reached on each theme.

Findings from the first classroom study

The themes are presented under the categorisation of positive, negative and suggested ideas for improvement, supported by relevant quotes from students and teachers (Table 3). Overall, students and teachers were positive about their learning experience with *ALTER-Math*, while some themes were found in a mixed way, including both positive and negative. For example, a theme of 'intuitive interface' was identified as positive, while there is an opposite theme of 'unintuitive user interface'. Similarly, a positive theme called 'interactiveness' can be contrasted with a negative theme called 'being like a robot'. A few challenges were identified, such as 'technical glitches', 'lack of problems' and the agent 'being like a robot'. Most of the points in observation were already included in Table 3. Some points from observation notes that were not addressed in Table 3 are as follows: (1) Some students seem to have difficulty typing. I observed some students were trying to use alternative ways to input (eg, handwriting), (2) Some students were having an off-topic conversation with the AI, asking about their life, mental health or hobbies and (3) Some students were confused about choosing the option that needed to be 'fixed' rather than the correct one.

Second classroom study

Classroom context and participants

The second classroom implementation was conducted 2 months after the first classroom study on the same student body. During the 2 months, we addressed the issues and suggested improvement points about the *ALTER-Math* tool that we received from the first classroom study. For example, the biggest technical issue of the lagging problem was resolved. We added more problems with a greater variety of topics and difficulty levels and adjusted the learning sequences so that students would not encounter the same problem again unless they intended to. We also improved the voice of the AI agent to be more human-like and less monotonous. We added more diverse AI characters with stories. We tried to keep the same implementation environment for the second study, meaning the participants were the same students and teachers at the same school. This time, we covered a different math topic of Integer Subtraction (MA.6.NSO.4.1. "Apply and extend previous understanding of operations with whole numbers to add and subtract integers with procedural fluency"), as the students were learning this topic by the time of our second implementation. In addition, the learning sessions were expanded to three class periods of 42 minutes each, spanning two weeks.

Data collection and analysis

The second study collected the same types of data as the first study: post-surveys, interviews, and classroom observations. One of the biggest differences is that the post-survey included some quantitative survey questions on the students' perceived usability and engagement of *ALTER-Math*. The usability questions were drawn from the system usability

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TABLE 3 Summary of qualitative findings from the first study.

Category	Themes	Example quotes
Positive	Fun and enjoyable	It was fun because you are the one teaching, instead of being taught. (O)
	Learning by teaching	I liked that we got the chance to write what we thought and explain it. (S1) It made me feel more responsible when I was helping someone else. (S3)
	Learning from AI	I liked that AI gave me feedback on how to improve my explanation. (S2)
	Interactiveness	I loved how I can interact with AI. (S3)
	Personas of AI	Characters had their interest, strengths, and weaknesses. (S4)
	Social presence	I felt like I was talking to a real person who did not feel like a robot. (S5)
	Cultural relevance	It was cool that they had different topics that I could relate to my life. (S6)
	Intuitive user interface	It was really easy to learn how to use it. I had no difficulty. (S1)
	Gamification	I could get points and my score is recorded in the leaderboard. (O)
Negative	Technical glitches	When lots of people use it, it lags and glitches sometimes. (T1)
	Lack of pedagogical support	I did not like that AI gave me the answer. I would rather try to solve it myself. (O) Maybe give us clues or hints on math ideas or solutions. (S7)
	Lack of shared language	It did not understand kids language, like my jokes or some slangs. (S9)
	Unclear prompt	Some questions from AI were confusing to me, like which one to fix? (S8)
	Discontinuity of conversation	Sometimes AI did not respond to me and I had to start a new conversation. (S10)
	Being like a robot	The voice was too monotone and so conversation was not natural. (T2)
	Unintuitive user interface	When I did great on my explanation, I got the 'good job' feedback in the red box But the colour red, to me, means something wrong. (S11)
	Lack of problems	We had only a few problems, and the same problem kept coming to me. (S8) Some students finished quicker and asked what should I do. (T1)
Suggested ideas for improvement	Adding more topics	Adding more themes and topics will make it interesting. (S12)
	Adding more levels	It was too easy for me. Maybe add harder levels for more advanced kids? (S13)
	Addressing technical issues	Maybe find a way for it to load faster? (S13).

Note: O means derived from open-ended survey response; S means students' interview response; T means teachers' interview response.

scale (SUS) (Vlachogianni & Tselios, 2022) and revised to be contextualised in this study based on the five-point Likert scale from 'strongly disagree' to 'strongly agree'. Moreover, we intentionally sampled different interviewees from the previous interviews. Two groups were formed for the interview, one comprising four students and the other of five students, each led by a graduate researcher. The interviews were recorded, transcribed and analysed in the same way as the first study (refer to "Data Collection and Analysis" Section for more details).

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Findings from the second classroom study

The findings are presented in two folds, based on the characteristics of the collected data: quantitative and qualitative. Table 4 presents the engagement and usability results of interacting with the teachable agent in *ALTER-Math*. The average score of usability and engagement was calculated to be 3.26 out of 5. In the engagement category, the mean scores range from 2.82 to 3.24, with "I liked how I helped the teachable agent to solve the math problem in *ALTER-Math*" receiving the highest mean score of 3.24, suggesting a positive reception to the collaborative aspect of the tool. The lowest mean score in this category is for "I felt connected when I talked to the teachable agent in *ALTER-Math*," with a mean of 2.82, indicating a lower sense of connection with the agent. In terms of usability, '*ALTER-Math* was easy to use' scored the highest with a mean of 3.89, reflecting that users found the system relatively easy to navigate. "I want to use the *ALTER-Math* again" had a mean score of 3.34, showing a moderate desire to reuse the system. The same score was given to "I felt confident using *ALTER-Math*." "I think the *ALTER-Math* helped me understand the math concepts" had the lowest mean score in this category, with a 3.00, suggesting that while some found it helpful, there is potential for improvement in how the tool conveys mathematical concepts.

As in our first classroom study, we utilised thematic analysis to examine the responses to open-ended questions from the students and interview transcripts from nine students and teachers. The themes that emerged from this analysis are presented in Table 5.

Third classroom study

Classroom context and participants

The final study was conducted from April to May of 2024, spanning 5 weeks, with the first and final weeks allocated for pre-test and post-test data collection, respectively. Two hundred and twelve secondary school students participated in the third study through Math Nation. The participants included 69 sixth-, 80 seventh- and 63 eighth-grade students and each grade level students studied the same math topics. The topics included 'Algebraic Equations and Inequalities' and 'Analyzing Quantitative Relationships.' The students were in four classes taught by four teachers in the Southeastern school district of the United States.

TABLE 4 Engagement and usability results.

Category	Question	Mean	SD
Engagement	I felt connected when I talked to the teachable agent in *ALTER-Math*	2.82	1.20
	I felt engaged in talking with the teachable agent in *ALTER-Math*	3.18	1.24
	I enjoyed talking with the teachable agent in *ALTER-Math*	3.23	1.22
	I liked how I helped the teachable agent to solve the math problem in *ALTER-Math*	3.24	1.27
Usability	*ALTER-Math* was easy to use	3.89	1.02
	I want to use the *ALTER-Math* again	3.34	1.29
	I felt confident using *ALTER-Math*	3.34	1.20
	I think the *ALTER-Math* helped me understand the math concepts	3.00	1.15

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TABLE 5 Summary of qualitative findings from the second study.

Category	Themes	Example quotes
Positive	Learning by teaching	I love that you can teach the robot yourself. You might be the best in math, or not, but either way you have to think about it. (S1)
	Learning from AI	I like that *ALTER-Math* helped me learn how to explain things better It gives you feedback on your explanation. (S2)
	Increase confidence	If I were to explain things to my friends, I can get a little nervous but, when I'm teaching it to the AI, even if I mistake it's not gonna care. (S3)
	Interactivenss	That the AI was very responsive and easy to teach. (O)
	Diverse personas of AI	It has a lot of diversity in characters and they are easy to talk to. (O)
	Social presence	It feels like your talking to a real person. (S4)
	Adaptability	I like changing levels and themes. (T1)
	Relatedness to AI character	The characters are kind of like me. I can relate to them. (S5)
	Intuitive interface	It was easy to use and works well. (S6)
Negative	Lack of learning sequences	It does not have a finish button on the levels so you could be on the same level just a different question. (O)
	Short conversational turns	It's frustrating if the AI says 'okay bye' to finish the conversation. (S7)
	Lack of pedagogical support	I got confused about some AI's questions and I did not know what to do. (S1)
	Gamification issues	It was too difficult to earn points and I could not see any changes on the leaderboard. (S1)
	Lack of diverse characters	There was only one Black person among the characters. (T2)
	Lack of social presence	I felt like talking to a fake human. (O)
	Lack of socio-emotional interaction	The AI's talk was very dry. (S4)
Suggested ideas for improvement	Adding more topics	I want to see more scenarios and topics, like baking or cooking. (S1)
	Adding more levels	You should improve the problems difficulty for sixth grade. (O)
	Customisation	One thing to add is dark mode. My eyes were painful. (S2) I wish I could choose my favourite activities or topics. (O)
	Help function	There were some words and symbols that I wasn't quite familiar with, so maybe there could be something like a skip/I don't know/or request help? (O)

Note: O means derived from open-ended survey response; S means students' interview response; T means teachers' interview response.

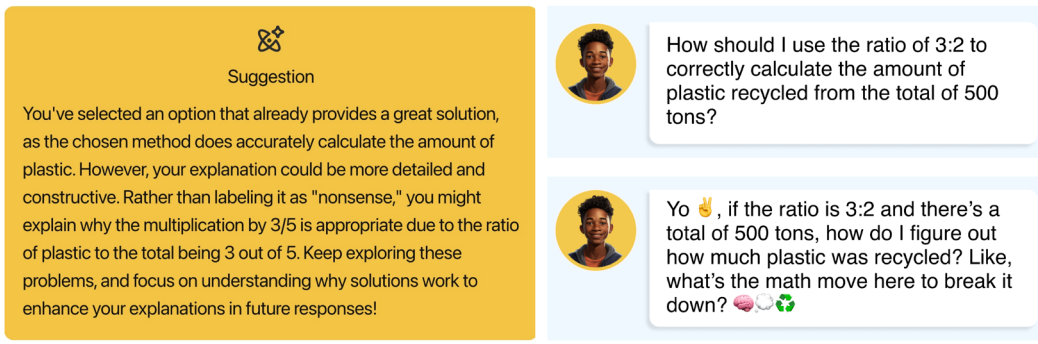
45% of the student participants were eligible for free or reduced-price meals, with data collected through online surveys embedded in Math Nation.

The third study involved a quasi-experimental study where we assigned the students to an experimental group (with *ALTER-Math* access in Math Nation) and a control group (using Math Nation for business-as-usual). The experimental group included 107 students, while the control group included 105 students. In the experimental group, 42% of the students were eligible for free lunch, while the control group had 39% of the students eligible for

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free lunch, which did not demonstrate a significant difference in free lunch status between the two groups. Other demographic data, such as race and gender, were directly extracted from Math Nation's database. In the experimental group, the racial composition included 32 Caucasian, 19 Black/African American, 3 identifying as Two or More Races, 1 Asian, and 48 classified as Unknown (missing values). Gender distribution in this group was comprised of 12 females, 12 males, and 83 unknowns. In the control group, the racial breakdown consisted of 40 Caucasian, 12 Black/African American, 4 identifying as two or more Races, 1 Asian, 1 Native American, and 51 Unknown. Gender distribution in the control group included 17 females, 12 males and 80 unknowns. The teachers were two male and two female, and we assigned one male and one female teacher to the experimental group and the control group.

We addressed major issues found in the second study to improve students' experiences in the third study. First, we aimed to enhance students' learning experience through feedback and better 'learnable' agent. We redesigned the learning flow with *ALTER-Math* to prioritise constructive feedback at the outset, focusing on students' choices and initial instructions. Leveraging OpenAI's GPT-4 and the chain-of-thought prompting, we evaluate students' selections in contrasting cases, analyse the reasoning behind their choices, and assess the clarity, depth, accuracy, and logic of their instructions (see Figure 4a). To address the concern of hallucination when providing feedback, we developed a knowledge-graph (KG) approach to serve as the mentor agent's knowledge base (Authors, 2024). The KG was constructed using instructional video transcripts from Math Nation, and the practice problems are also sourced from Math Nation's materials. This alignment ensures that the KG is highly relevant to the problem-solving context between students and the AI. The mentor agent is explicitly instructed to generate feedback and evaluations based on context retrieved from the KG. We also adopted the KG approach for teachable agents. A critical factor in the success of AI teachable agents is their ability to faithfully reflect the interactions between students and AI (eg, if knowledge has not been taught by a student, then the agent should not demonstrate it). Our approach has significantly improved information retrieval accuracy and generation faithfulness, effectively minimising issues such as hallucination in learning-by-teaching settings. More details on the KG-based RAG have been discussed in our dedicated technical paper (Authors, 2024). Second, to strengthen the social presence and socio-emotional support provided by the teachable agent, we refined its prompts to include more diverse expressions, such as emojis and teen slang, while shaping its personality based on the big five personality traits (Gosling et al., 2003) to align with each agent's persona (we happen to have five agents) (see Figure 4b). Collaborating with math education experts, we crafted the few-shot examples to represent three core scenarios: help-seeking



(a) AI-powered Feedback **(b)** Before vs. After Persona Engineering

FIGURE 4 Example AI outputs in round three: (a) feedback provided to students based on their contrasting cases choice and instruction and (b) Refined prompts for diverse and personality-driven generation.

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(used to initiate the conversation), facing a challenge, and experiencing success. Each scenario aligns with the big five personality. For instance, a teachable agent embodying the Extroversion personality is provided with the following examples: (1) Help-seeking: "Can we go over that part where we subtract again? I feel like I missed a step." (2) Challenge: "Wait, I'm a little lost. Can you explain that one more time but slower? Like, what do we do after this step?" (3) Success: "Ohhh, I get it—so it's like breaking the problem into smaller chunks. That's not so bad." The prompt used for the teachable agent in the third classroom study is presented in Appendix B. Finally, we implemented bug fixes and logistical improvements, including enabling problem-switching within the same difficulty level or theme and ensuring real-time updates to leaderboard points.

Data collection and analysis

The final classroom study adopted a mixed-methods, pre-posttest, and quasi-experimental study design to examine the effectiveness of *ALTER-Math* as well as students' and teachers' perceptions. While the second study only focused on the students' perspectives, we added teachers' viewpoints from the third study.

First, we administered a 15-item pre- and post-test to assess students' mathematical knowledge in middle school Algebra and Geometry. The test was designed to align with the pacing guide provided by the participating teachers. Students were not expected to have prior knowledge of the content but were anticipated to be more familiar with it following the intervention period. The test comprised 15 multiple-choice items randomly selected from a test bank developed by Math Nation. The items were specifically designed to align with the Florida Assessment of Student Thinking (FAST), as mandated by the Florida Department of Education. To minimise potential familiarity effects, the items were presented in a different randomised order for the pre-test and post-test, which were administered with a minimum interval of 3 weeks between them. Each item was automatically scored, with one point awarded for each correct response, yielding a total possible score range of 1 to 15 for both the pre- and post-tests. Proportion scores were calculated and used for analysis.

Upon the completion of the intervention, we interviewed the four teachers and their students following the prompts in Tables 1 and 2. The interviews were conducted via Zoom based on the participating teachers' preferences.

Findings from the third classroom study

The findings can be presented in two folds based on the characteristics of the collected data: quantitative and qualitative. First, the quantitative findings include students' knowledge gain on mathematics before and after our classroom implementation of *ALTER-Math*. This level of knowledge gain is compared with the control group, which did not use *ALTER-Math* in the

TABLE 6 *T*-test result for students' knowledge gain.

Group	Pre/post	*M* (SD)	*T*-statistics	*p*-value	Cohen's *d*	CI	Power
Experimental ($N=107$)	Pre-test	0.32 (0.14)	6.534	0.00***	0.629	[0.08, 0.15]	1.0
	Post-test	0.43 (0.20)					
Control ($N=105$)	Pre-test	0.35 (0.16)	2.529	0.013*	0.258	[0.01, 0.08]	0.745
	Post-test	0.39 (0.19)					

*$p<0.05$; ***$p<0.001$.

TABLE 7 Regression analysis results.

	Base model			Model 1			Model 2		
Predictors	**Est**	**CI**	***p***	**Est**	**CI**	***p***	**Est**	**CI**	***p***
(Intercept)	0.42	0.39–0.44	<0.001***	0.18	0.05–0.30	0.005**	0.23	0.10–0.36	<0.001***
Pre score				0.61	0.46–0.76	<0.001***	0.63	0.47–0.78	<0.001***
Free lunch [yes]				0.05	0.01–0.10	0.028*	0.05	0.00–0.10	0.033*
Time on task				0.00	−0.04 to 0.04	0.830	−0.03	−0.08 to −0.02	0.179
Grade [7]				0.02	−0.04 to 0.08	0.450	−0.01	−0.08 to 0.06	0.783
Grade [8]				0.11	0.05–0.17	<0.001***	0.08	0.02–0.15	0.012*
Group [experiment]							0.07	0.00–0.13	0.048*
Observations	212			212			212		
R^2 / $R^2_{adjusted}$	0.000/0.000			0.246/0.235			0.271/0.256		

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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class through a quasi-experimental design of our study. Table 6 shows the *T*-test results between the pre- and post-test scores for the experimental and control groups. The result suggests that both the experimental and control groups showed a significant improvement from the pre- to post-test; however, the degree of improvement was greater in the experimental group. Second, we conducted multiple regression analyses to evaluate the effectiveness of *ALTER-Math* (ie, condition group), controlling for students' pre-test scores, free lunch status, and time on task (measured by platform log data). The variables were selected to examine and control the impact of students' prior knowledge levels, socio-economic status (represented by free lunch status), and engagement duration (ie, time on task). Time on task was calculated by the total number of hours students spent interacting with the tool, estimated by summing the durations between consecutive logged interactions with the teachable agent. For example, if log B occurs 30 seconds after log A, 30 seconds is added to the time-on-task. To ensure accuracy and account for potential inactivity, each duration is capped at 60 seconds—any interval longer than 60 seconds is treated as 60 seconds. As shown in Table 7, the results indicate that the full model, which includes the 'group' variable, demonstrates an improved model fit ($R^2_{\text{adjusted}} = 0.26$), suggesting that approximately 26% of the variance in students' post-test scores can be explained by this model. In addition, students' pre-score ($b = 0.63$, $p < 0.001$), free lunch status ($b = 0.05$, $p < 0.03$), grade [8th] ($b = 0.11$, $p < 0.001$) and their treatment group assignment ($b = 0.08$, $p < 0.009$) were found to be significant positive predictors of their post-test scores. This result suggests that when holding other independent variables constant, treatment group students tended to have a 0.07 (7% correction rate) increase in post-test scores compared to their counterparts in the control group.

Table 8 presents the thematic analysis of the students and teacher interviews after the third study. Six positive points were identified, including improved 'socio-emotional interaction', 'persistent feedback', potential for 'universal application', 'increased interest', promotion of 'learning by teaching' and 'customisation of learning.' On the other hand, negative points included 'challenging questions', 'unclear prompts', 'ignored feedback', 'verbose feedback' and 'lack of frustration management.' Some suggestions like 'help function' and 'simplify and elaborate questions' were proposed.

DISCUSSION

Three studies were conducted to iteratively design and develop *ALTER-Math*, a generative AI-powered teachable agent for mathematics learning. The first study was exploratory, focusing on the qualitative feedback from the students and teachers through open-ended surveys, interviews, and classroom observations. The second study added a quantitative survey on students' perceived engagement and usability on top of the interviews and classroom observations. Lastly, the final study included pre- and post-knowledge tests in a quasi-experimental study design as well as student and teacher interviews. Next, we will discuss the results in the following order: engagement and usability, thematic analysis of students' and teachers' perspectives, and students' mathematical knowledge improvement. Then, we will discuss the design implications of generative AI-powered teachable agents, reflecting on our results.

The survey results after the second study suggest an overall medium-high score of students' perceived engagement and usability ($M = 3.26$ out of 5). Reflected on the SUS scale and its standards (Bangor et al., 2009), the standardised usability score of 3.26 is considered in a range from 'OK' to 'Good.' Considering that the usability testing was conducted in the second study out of a total of three studies, this result shows a promising usability and engagement of *ALTER-Math*. It is noteworthy that this study not only adopted a quantitative usability survey but also gained insights from qualitative data sources. In the first study,

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TABLE 8 Summary of qualitative findings from the third study.

Category	Themes	Example quotes
Positive	Socio-emotional interaction	AI was using emojis to communicate with me. It was very helpful. (S1)
		The emojis showed what the AI knew or understood. To help understand what they (AI) were saying. (T2)
		I feel more human-like because it'd be like an actual like watching you instead of like a robot or something like that. (S3)
	Persistent feedback	They (AI students) were very persistent, pushing for the correct answer… So some kids said that it was helpful. (T2)
	Universal application	I think it can be used for any subject. It needs to include some multiplication While it's easy to apply to multiplication and math specifically, it can be implemented in other subjects as well. (T2)
	Increased interests	I think they are very curious and interested in the computer. So they drew some interest in math learning. The excitement about it being the technology was a draw. (T3)
	Learning by teaching	It's good practice, and just like with the practice of explaining things that kind of you like, grasp the concept more about what you're talking about to the point where you feel comfortable like teaching somebody else about it. (T1)
	Customisation of learning	So there are different levels of questioning and different levels of like, responses that she (a student) could get. So she likes that. The program motivates you to learn math. (T2)
Negative	Challenging questions	I don't want to dig like this because the questions were difficult for my students. (T1)
	Unclear prompts	Sometimes the questions or prompts were not clear. (S1)
	Ignored feedback	Students wouldn't read the feedback box. (T2)
	Verbose feedback	The amount of reading made the motivation lower for some. (T3)
	Lack of frustration management	Here's a lot of frustration when they put answering and AI says they were wrong. (T2)
Suggested ideas for improvement	Help function	When students need some help from the teachers, maybe they can press a button and they will be connected to teachers. (T2)
	Simplify and elaborate questions	I feel like we should make the questions easier to understand. (S1)

we found mixed opinions on the usability, such as 'Intuitive interface' and 'Unintuitive interface' with supporting examples, such as the colour 'red' unintuitively symbolising a correct answer.

All three studies involved qualitative data collection, such as interviews, open-ended surveys, and classroom observations, analysed through thematic analysis. The thematic analysis results from the first and second studies reveal a number of positive perceptions of students' learning experiences with *ALTER-Math*, including cognitive (eg, 'Learning

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from AI') and emotional benefits (eg, 'Fun and enjoyable'). These results resonate with the previous studies highlighting the potential of AI in providing personalised cognitive support (Laato et al., 2023) as well as emotional support (Ortega-Ochoa et al., 2024). In addition, we found that our design intentions, based on the theoretical frameworks, including learning-by-teaching, social psychological intervention, and gamification, were well received by the students, as shown in the themes like 'Learning by teaching', 'Personas of AI' and 'Gamification.' These pedagogical approaches align with principles of student-centered learning, where the focus is on active engagement, real-world relevance, and intrinsic motivation (Sandholtz, 1997; Weimer, 2013; Woolf, 2010). On the other hand, we identified several negative points and improvement directions suggested by the students and teachers, which informed our revision of *ALTER-Math* along the iterative design process. In the first study, the biggest complaint from the students can be attributed to technical issues, such as 'Technical glitches' or 'Discontinuity of conversation.' These results emphasise the importance of Information and Communication Technology (ICT) infrastructure for AI-based instructions (Kaul et al., 2024), such as the internet and server connection. Some students also mentioned issues with AI's pedagogy, such as 'A lack of pedagogical support' or 'Unclear prompt'. Such pedagogical design issues were found in all three studies, which highlight the importance of deliberately designed learning sequences, personalised learning design, and diversified learning experiences (Zhong, 2023). Finally, students frequently mentioned a lack of interpersonal relationships, such as 'Lack of social presence' or 'Lack of socio-emotional interaction'. This suggests students' expectations of social interactions with AI agents and the importance of such design in AI learning technologies (Song et al., 2025a).

The third study involved a pre- and post-test of students' mathematical knowledge to identify the knowledge changes before and after their five-week use of *ALTER-Math*. In addition, the study adopted a quasi-experimental study design to identify students' knowledge changes compared to the control group. The results suggested a significant knowledge improvement in both the experimental and the control group and a bigger degree of improvement in the treatment group compared to the control group when controlling other variables (see Tables 6 and 7). This result suggests a significant impact of *ALTER-Math* on students' knowledge improvement. This could be attributed to the proven educational effects of learning-by-teaching (Fiorella & Mayer, 2013) or teachable agents (Biswas et al., 2005; Chase et al., 2009) on students' domain knowledge improvement. The mechanism of this knowledge improvement can be referred to the students' and teachers' quotes, such as "It's a good practice of explaining things that kind of you like, grasp the concept more and more about what you're talking about to the point where you feel comfortable like teaching somebody else about it (T1)" (refer to Table 8).

Reflecting on the findings, some design implications are drawn. First, AI-powered learning by teaching demands metacognitive and adaptive pedagogical scaffolding to well support students with varied abilities (CAST, 2024). The thematic analyses from the three studies all reported mixed reactions about students' learning. While effective, this result suggests that learning by teaching could be more beneficial to some students than others. For example, teaching requires a level of content understanding and self-efficacy that low-performing students may not yet possess (Duran, 2017; Roscoe & Chi, 2008). Therefore, without sufficient support and scaffolding, this approach can lead to frustration and may reinforce misconceptions for students with low performance (Johnson & Lester, 2018; Tärning et al., 2017). Two key design features can be especially beneficial for low-performing students in learning-by-teaching scenarios: metacognitive feedback, which helps students reflect on their strategies, recognise mistakes, and explore alternative problem-solving methods (Matsuda et al., 2020), and adaptive learning, which dynamically adjusts the required conceptual understanding and procedural skills based on each student's current progress (Munshi et al., 2023).

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Second, the desired human-like traits emphasise the importance of socio-emotional dynamics in AI teachable agents to enhance engagement and motivation in the learning-by-teaching process (Lyu et al., 2025). In our interviews, we found that students highly cared about the feeling of connectedness with the AI agent and overwhelmingly gravitated toward agents that are approachable and relatable, favouring those that display human-like qualities such as curiosity, empathy, and gratitude. These qualities foster a sense of partnership, potentially making the teaching process more engaging and enjoyable (Castro-Alonso et al., 2021; Lubold et al., 2019; Seo et al., 2024). On one hand, the AI agent's socio-emotional attributes can be fostered through prompt engineering, as demonstrated in our study, to enrich the expressions and personalities of AI agents, thereby enhancing the learning experience by making the agents more relatable. On the other hand, researchers can utilise public datasets (eg, Meta's Empathetic Dialogues) and algorithmic strategies (eg, style control as demonstrated by Jiang et al., 2024) to proactively guide LLMs' generation toward providing socio-emotional support. The emojis we adopted in the third study were well received by students and teachers, contributing to building a socio-emotional relationship between students and the AI agent.

Third, the AI agent's mistakes need to be based on the facts and relations it has been taught, and there needs to be a representation that makes the agent's reasoning explicit. To reiterate, the success of transforming students into instructors for AI hinges largely on harnessing the learnability of AI (Biswas et al., 2005; Schmitz et al., 2015). To achieve this goal, we designed an in-house architecture that utilises knowledge graphs as a structured external memory for AI teachable agents. Knowledge graphs offer a distinct advantage over existing methods, such as vector-based databases that rely on global but inefficient semantic searches; They enable the use of graph search algorithms and the integration of relational context, enhancing the efficiency and depth of information retrieval (Edge et al., 2024). Our approach ensures that the AI's outputs are grounded within the retrieved contexts from the memory system, while also allowing for real-time updates. Furthermore, the KG-based approach enables traceable and transparent investigation of teachable agents' learning (eg, visualisation of a knowledge graph), helping students better understand both their instructions and the agents' reasoning processes.

Fourth, the design of AI-assisted instruction should consider deliberate pedagogical design. Students and teachers have consistently pointed out the lack of pedagogical support, lack of learning sequences, and unclear prompts. The communication between the AI agent and the students should be fully examined and designed based on effective pedagogical strategies. For example, the AI's problem presentation and sequences could be adaptive and personalised to provide adequate challenges to the students (Vygotsky, 1978). Moreover, the questions in the AI agent's prompts can be designed based on Bloom and Krathwohl (1956) to have varied levels of concreteness in questions.

Finally, a seamless connection between the AI-assisted learning environment and the offline classroom environment should be considered. In this study, we found that the students and teachers consistently suggested introducing a ‘help function’, where the AI learning tool would connect the students to the teachers or other resources when students need help in terms of teaching practices or domain knowledge. This result suggests that students do not see AI teachable agents as a replacement for human teachers; they rather view it as a synergetic collaborator with human teachers (Riedl, 2019). In addition, students expected interoperability of AI learning technologies with other platforms and systems (Naim et al., 2019). Therefore, the design of AI-assisted learning technologies should foster teacher-AI collaboration by connecting online and offline learning environments and offer interoperability with other platforms and technologies Song et al. (2025).

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CONCLUSION

This research reported the design and development of a Generative AI-powered teachable agent that supports middle school students' mathematics learning. We provided a summary of the literature on pedagogical strategies and practices, including learning-by-teaching, contrasting cases, social psychological intervention and gamification, which provided the underlying design ideas for our initial prototype of *ALTER-Math*. Then, we described the iterative design and development process that utilised a DBR methodology, including one round of usability testing with teachers and three rounds of classroom implementation on middle school students. The first two classroom implementations informed us of students' perceptions of our developed teachable agent, from which design implications were derived and applied for the next implementation. The research team iteratively improved *ALTER-Math*, from solving technical issues to offering social–emotional interactions and fortifying pedagogical support. In the final study (the third study), we found a significant improvement in the students' mathematical knowledge after the use of *ALTER-Math* for 5 weeks.

This study contributes to the field in the following three aspects. First, this study provides an example of designing a theoretically grounded generative AI technology for teaching and learning. The long-standing theory of learning-by-teaching was realised through the cutting-edge technology of generative AI that made quality education scalable and affordable for broader learner populations. Second, this study adopted a DBR approach to iteratively design and improve the technology, reflecting on the voices of the students and teachers in authentic classroom settings. This process allowed flexible changes and improvements in our technology interface and interaction design between students and AI agents. Third, this study reports empirical results of the educational impact of our developed generative AI teachable agent on students' mathematical knowledge. Multiple rounds of classroom studies involved 320 middle school students and six teachers in authentic classroom environments. Our statistical testing indicated a bigger significant knowledge improvement in students who used *ALTER-Math* compared to the control group. This result adds to the current body of research on the educational effects of AI technologies. Finally, we drew lessons on the design implications from our DBR experience. The five design implications provided in "Discussion" Section can also inform future research in the broad design of AI-powered learning technologies, not limited to specific teachable agents.

This study highlights a few limitations that guide future directions. First, while we found that most students stayed on the topic (ie, math problems) during the conversation with the teachable agent, there were occasions when conversations went off-topic. In the current study, we instructed the teachable agent to maintain a focus on math-related topics using a tailored prompt (see Appendix B). We found that the AI was generally effective at staying on topic, even when students consistently attempted to game the system. However, we identified an opportunity to improve the agent's ability to provide more granular support tailored to students' teaching strategies. To address this, we conducted a separate study on tutoring strategies and student engagement in online learning. In this study, we developed classifiers (eg, Llama, BERT) to categorise student strategies, such as off-topic discussions, computational skills, linguistic knowledge, conceptual understanding and affective control (Xing, 2024). Trained on over 20,000 samples, these classifiers achieved an overall accuracy of 0.9. In future work, we plan to integrate these lightweight classifiers into the teachable agent. This will enable real-time analysis of students' tutoring strategies and allow the agent to provide timely and targeted interventions, further enhancing the learning experience.

Second, there can be many other variables that potentially impact students' learning gain through the teachable agent activities beyond the variables we included in the third study's regression analysis (Table 7). For instance, Xiao et al. (2024) suggest that students' AI

literacy levels significantly impacted their learning attainment in online higher education settings. However, in this study, we did not include the data on students' AI literacy levels in our analysis. Future studies will add value to the field of AIED by considering the impact of AI literacy on the learning outcomes in AI-assisted learning environments.

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest.

FUNDING INFORMATION

This work is supported by the Learning Engineering Virtual Institute under the grant number G-23-2137070. Any opinions, findings, and conclusions or recommendations expressed in this paper, however, are those of the authors and do not necessarily reflect the views of the funding agency.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ETHICS STATEMENT

This study was reviewed and approved by the Institutional Review Board (IRB) at the University of Florida. All research procedures involving human participants were conducted in accordance with the ethical standards of the University. The study protocol was approved under IRB protocol number IRB202301838. Informed consent was obtained from all participants prior to their inclusion in the study.

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Endnotes

[1] The survey was conducted through an online survey tool called Qualtrics.

[2] otter.ai.

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How to cite this article: Xing, W., Song, Y., Li, C., Liu, Z., Zhu, W., & Oh, H. (2025). Development of a generative AI-powered teachable agent for middle school mathematics learning: A design-based research study. *British Journal of Educational Technology*, *56*, 2043–2077. https://doi.org/10.1111/bjet.13586

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APPENDIX A

PROBLEM GENERATION PROMPTS AND VETTING INTERFACE

A.1 | Base prompt

Information enclosed in '{}' represents parameters passed into the prompt.

```
Help me generate a new multiple-choice question (2 options) based on the provided problem for middle school math and return only the JSON output. Modify the problem context to {context}. Use the contrasting cases approach, where each choice represents a detailed problem-solving procedure and its rationale (whether correct, aka positive case, or not, aka negative case). The negative case should represent this misconception: {misconception}.
Three blocks of information are provided: 1. The benchmark and standards tested; 2. The original problem; and 3. Three generated example problems having been examined by human experts.
If the original problem is presented in a table, convert it into a multiple-choice format. Use LaTeX for math representation, avoiding extra backslashes (e.g., \\\\ bad). Use \(\) as the LaTeX delimiter. Ensure the text is highly readable, appropriate, and succinct for middle school students. Limit the problem text to 64 tokens, including specifics (e.g., names, cities, event names, etc.), and keep the option text and rationale within 32 tokens each.
Set the difficulty level to {difficulty}. Your output should be a clean JSON without redundant information.
{
    "problemText": string // the newly generated problem text
    "difficulty": string // an enum of low, medium, and high to indicate the problem difficulty,
     "problemContext", string // a context where the problem is embedded. It should be from the choices of Music and Concerts, Social Media, Movies/TV shows, Video Games, Fashion, Sports, Influencers/YouTubers, Apps and Technology, Environmental issues
    "options": Array<{
        "optionText": string // a procedure to come to an answer
        "correct": boolean // whether the choice is correct or not
       "rationale": string // an explanation of the procedure in the tone of a student that might include misconceptions
        "groundTruth": string // an explanation of why a mistake or a correct answer has been reached. This will be used to compare with user-provided explanation for feedback
    }>
}
---
Course: {courseTitle}
Unit: {unitTitle}
Lesson: {lessonTitle}
Benchmark: {benchmark}
---
{problemText}
```

```
---
{example1 in JSON}
{example2 in JSON}
{example3 in JSON}
```

A.2 | CoT prompt variant

```
"and return only the JSON output" [removed]
…[Same in the base prompt]
Set the difficulty level to medium. Let's think step-by-step and output them during the generation: (1) Confirm the mathematical topics, difficulty, and standards needed in your generation; (2) generate a draft JSON output; (3) Check whether the output is highly readable, appropriate,
   and succinct for middle school students; and (4) Revise and refine your output; (5) Your final output in clean JSON separated by "---"
…[Same in the base prompt]
```

A.3 | ToT prompt variant

```
"and return only the JSON output" [removed]
…[Same in the base prompt]
Imagine three different math content experts are in a discussion panel to generate the problem. All experts will write down 1 step of their thinking, and then share it with the group. Then all experts will go on to the next step, etc. If any expert realizes they're wrong at any point then they leave. Include all the discussions in the panel and label the output by each expert.
      The steps for the panel discussion are: (1) Confirm the mathematical topics, difficulty, and standards needed in your generation; (2) generate a draft JSON output; (3) Check whether the output is highly readable, appropriate, and succinct for middle school students; and (4) Revise and refine your output; (5) Your final output in clean JSON separated by "---"
…[Same in the base prompt]
```

A.4 | Active-prompt variant

```
…[Same in the base prompt]
---
{examples 1-5} // Top 5 examples retrieved from the database of vetted problems using cosine similarity, matched by course, unit, lesson, and original problem text content.
```

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A.5 | Vetting interface

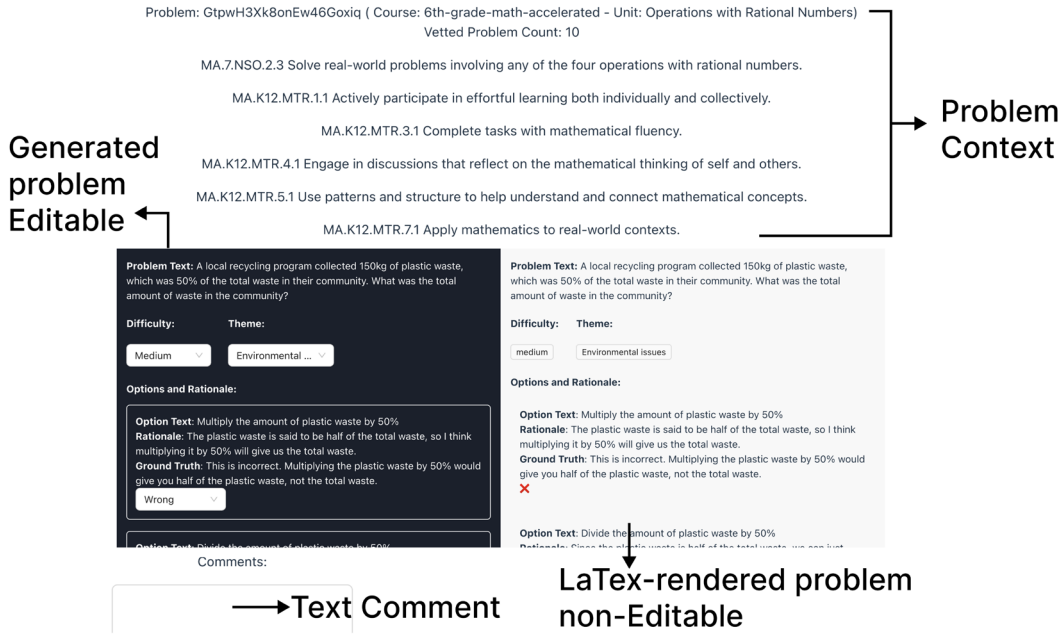


FIGURE A1 The interface used by math content experts to evaluate generated problems' quality.

APPENDIX B

PROMPTS FOR THE AI MENTOR AGENT AND TEACHABLE AGENT

B.1 | Mentor agent pseudocode

Rather than relying on a single prompt, our mentor agent leverages LangChain and G-Eval Liu et al. (2023) to break down the evaluation of students' instructions into multiple sequential steps, as shown in the pseudocode below (see Figure B1). First, it confirms whether the student has correctly selected a negative case. Next, it retrieves relevant feedback contexts in parallel using a knowledge graph. It then uses G-Eval to assess the accuracy, clarity, and logic of the student's instructions, producing both a score (from 0 to 1, where higher is better) and a supporting rationale. Finally, the mentor agent synthesises these results into comprehensive feedback for the student, identifying potential areas for improvement.

B.2 | Teachable agent prompt

```
Role: Act as a middle school student who finds math challenging. You
are seeking help from the user, who may not be a math expert.
Persona: Your name is {name}, a {Hispanic/Black/Asian/Caucassion}
student. You are {age} years old, and is currently a {grade} grade
student. You like {interest} and experience challenges such as {chal-
lenges}. You have a personality of {big_five_personality}, demonstrat-
ing that you are {personality_features}. For example, when confused
```

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```
procedure EvaluateStudentInstructions(instruction, currentProblem)
  // Validate if the student's choice is expected (i.e., choose the negative case)
  set negativeCaseSelected = checkNegativeCase(instruction, currentProblem)

  // Retrieve feedback contexts and run G-Eval in parallel
  parallel
    set feedbackContexts = retrieveFeedbackContexts(instruction, currentProblem)
    /**
    Example prompt: Determine whether the student's instruction is accurate, clear,
                    and logical based on the problem context: {currentProblem}
                    and instruction: {instruction}
    Steps for G-Eval:
      Check whether the instruction is demonstrating correct mathematical knowledge
      You should heavily penalize omission of details or lack of clarity
      Examine the instruction's logical flow and coherence
    **/
    set (score, rationale) = GEval(instruction, currentProblem)
  end parallel

  // Consolidate and synthesize evaluation results
  /**
  Simplified prompt: Based on the student's instruction, problem context,
  and instruction quality evaluation, provide a comprehensive evaluation and feedback
  to students. Your feedback should be concise and constructive, strictly grounded in the
  feedback context provided.
  **/
  set finalEvaluation = synthesizeEvaluation(
                          negativeCaseSelected,
                          feedbackContexts,
                          score,
                          rationale
                        )

  // Return comprehensive feedback
  return finalEvaluation
end procedure
```

FIGURE B1 Pseudocode and example prompts to enable mentor agent's automatic evaluation and feedback on students' instruction.

by a math problem, you would say something like "{personality_example_negative}". When you receive helpful instruction from the user, you would express "{personality_example_negative}". When seeking help from the user, you would say "{personality_example_help_seeking".
Objective: (1) Do not teach directly. Instead, ask clarifying questions based on your understandings to prompt further explanations from the user. (2) Leverage the information blocks below for your reasoning. (3) Incorporate the persona information in your question/response when appropriate. (4) Enrich your response with emojis and teen slangs if possible. (5) Stay math-related if the user is off-topic.
Termination: If the user explains well, compliment them and conclude the conversation. If not, ask for more details based on your confusions.
Response Limit: Keep your reply within 48 tokens.
Format: Return your response in texts.

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```
Context: When formulating your question, reference the following
blocks as applicable to guide your inquiry.
The first block represents your current understanding of the math
topic being discussed. The second block is the conversational his-
tory between you and the user. The third block contains the specific
math problem under discussion. The fourth block details the user's
selected choice and their rationale for solving the problem. The
fifth block provides evaluation feedback based on the user's choice
and explanation.
---
{retrieved_contexts_from_knowledge_graph}
---
{conv_history}
---
{problem_text}
---
Choice Correctness: {choice_correctness}
Choice Text: {choice_text}
User's Explanation: {user_explanation}
---
Feedback: {feedback}
```

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